

CANDIDATE NAME (CG) _____

INDEX NUMBER _____



SERANGOON JUNIOR COLLEGE
JC2 Preliminary Examination 2016

BIOLOGY
Higher 2

Paper 3

19th SEP 2016/ Monday
2 hours

MARK SCHEME

READ THESE INSTRUCTIONS FIRST

Write your name and index number in the spaces at the top of this page and on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in all the sections.

At the end of examination,

1. fasten all your work securely together

FOR EXAMINER'S USE	
1	/14
2	/14
3	/12
	/40
4	/12
5	/20
TOTAL	/72

INFORMATION FOR CANDIDATES

The intended number of marks is given in brackets [] at the end of each question or part question.

This question paper consists of 17 printed pages and 1 blank page

Answer **all** questions

Question 1

The first ever gene therapy trial was initiated in 1990 by Dr William French Anderson to treat a four year old girl named Ashanthi. Ashanthi was suffering from severe combined immunodeficiency (SCID), a genetic disorder characterised by the absence of functional T-lymphocytes.

In Ashanthi's case, the disease was caused by the absence of the enzyme adenosine deaminase (ADA-SCID). An alternative form of SCID is known as X-linked SCID.

(a) From your knowledge, contrast between the two forms of SCID mentioned in the paragraph above. [3]

	X-linked SCID	ADA-SCID	
Gene involved	Gene coding for interleukin 2 receptor gamma	Gene coding for adenosine deaminase	1m
Chromosome on which the gene is found	X chromosome	Chromosome 20	1m
Effect of mutation leading to SCID	Results in production of defective interleukin receptor	Mutation results in non-functional enzyme that is defective in purine metabolism / cannot break down purines	1m
	Leading to defective signalling pathway which prevents proper development of T-lymphocytes	Leading to the accumulation of deoxyadenosine which is toxic to immature lymphoid cell	1m

[max 3]

There were several *ex vivo* methods of gene therapy that were considered in Ashanthi's treatment, detailed in Table 1.1. The virus vector used was a modified retrovirus.

Method	Description
A	• Normal ADA allele is introduced into viral vector • Recombinant virus is introduced into T-cells obtained from patient • Genetically modified cells are reintroduced into patient
B	• Normal ADA allele is introduced into viral vector • Recombinant virus is introduced into the cells derived from inner cell mass of blastocyst • Genetically modified cells are reintroduced into patient.
C	• Normal ADA allele is introduced into viral vector • Recombinant virus introduced into hematopoietic stem cells obtained from patient • Genetically modified cells are reintroduced into patient

Table 1.1

(b) With reference to the information presented in Table 1.1,

(i) Suggest one reason why *ex vivo* approach was utilised for gene therapy. [1]

- Allows specific **identification of target cells** with therapeutic gene (OWTTE).
- Allows **monitoring of gene expression before reintroduction** as success rate is very low.
- Safer because can **monitor for cancer cells** caused by the **random integration of DNA / insertional mutagenesis by retrovirus** which can knock out tumour suppressor genes or activate proto-oncogenes.

[any 1]

(ii) State the most preferred method of *ex vivo* gene therapy for ADA-SCID and justify the preference over the other 2 methods. [3]

- Method C.
- **Hematopoietic stem cells** are capable of **long term self renewal**, making the treatment **more permanent** as compared to using **differentiated T-cells** in **method A**.
- **Hematopoietic stem cells** are derived from the patient **will not trigger any immune response** when returned to patient, as compared to using **embryonic stem cells** from another source in **method B**.

(c) Explain why the retrovirus was an efficient vector in the gene therapy of Ashanthi's condition. [2]

- It contains surface **glycoproteins** which are **specific** to the cell-surface receptors on T-**lymphocytes**, thus **increasing the efficiency** of gene delivery to the **specific target cell**.
- It contains the enzyme **integrase**, allowing the **integration** of the **normal ADA allele** into the **genome** of the target cell.

Apart from diseases that plague human health, the effect of plant diseases on agriculture have also been in the spotlight. One such disease is the ringspot virus that plagues the papaya agricultural industry. Scientists have developed effective circumventive methods to tackle the problem of the ringspot virus through genetically modifying papaya. To do this, viral genes encoding capsid proteins were transferred to the papaya genome. These viral capsid proteins elicit something similar to an "immune response" from the papaya plant. Thus, the genetically modified papaya plants were resistant to infection by the papaya ringspot virus.

Figure 1.2 below depicts the comparative infection of transgenic and non-transgenic papaya in the 1995 field trial in Kapoho, Hawaii.

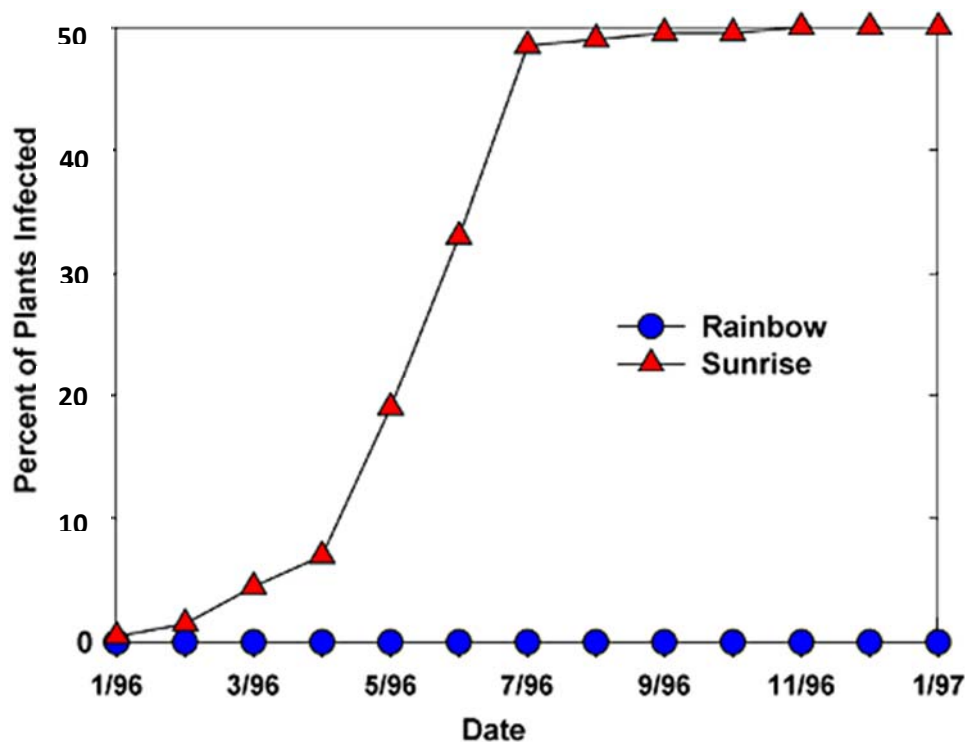


Figure 1.2

(d) With reference to Figure 1.2,

(i) Determine the identities of the transgenic and non-transgenic papaya species. [1]

- Transgenic species: Rainbow + Non-transgenic species: Sunrise.

(ii) Evaluate and justify thoroughly the efficacy of the genetic intervention. [2]

- Highly effective
- Quote values:

The percentage of infected Sunrise papaya species increased to from 0% to 16% by April'96, followed by a sharp increase from 16% to 98% by July'96 and reached 100% by Nov'96, as compared to the percentage of infected Rainbow species which remained at 0% throughout the year.

It was observed that three years later, the percentage of infected transgenic papaya species increased.

(e) Suggest and explain a possible reason for this phenomenon. [2]

- Mutation in the ringspot virus genome had occurred,
- Enables ringspot virus to make new surface glycoproteins that GM papaya is no longer resistant to as immune response is no longer triggered by infection with the ringspot virus.
- AVP+ explanation

[Total: 14 marks]

Question 2

The location of the gene locus responsible for disease X was not discovered until 1985. Scientists used restriction fragment length polymorphism to discover the genetic markers associated with the disease. One such marker was a 950bp-long region known as XD15 that had been sequenced prior to 1985.

Samples of DNA were obtained from a family known to have the condition. The XD15 locus was amplified by polymerase chain reaction and mixed with *Pst*I and *Eco*RI in two separate restriction digests. The results of gel electrophoresis followed by southern blot of both restriction digests are shown in Fig. 2.1.

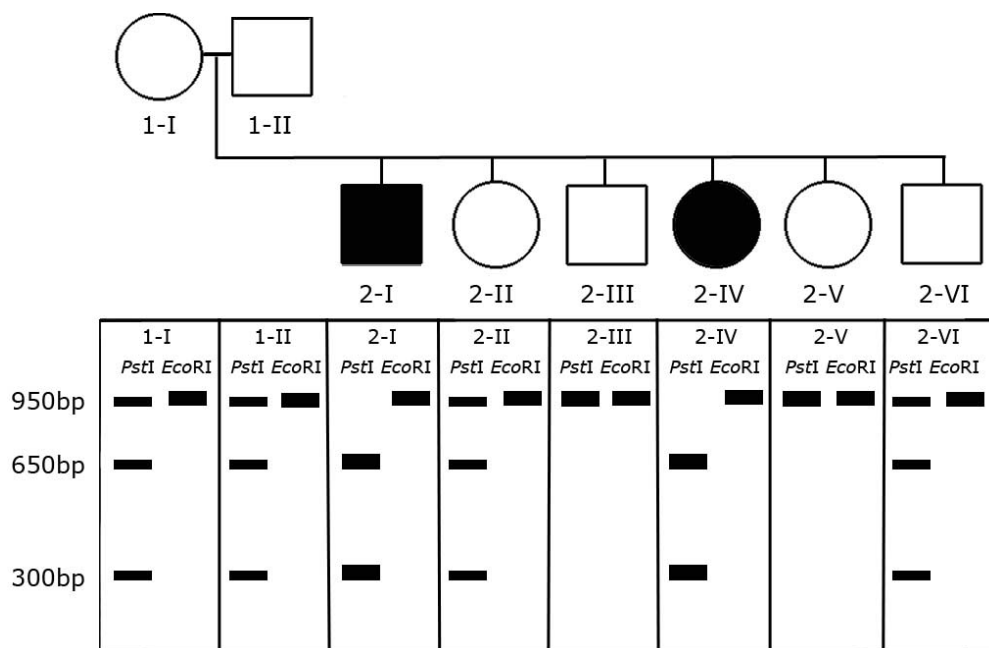


Fig. 2.1

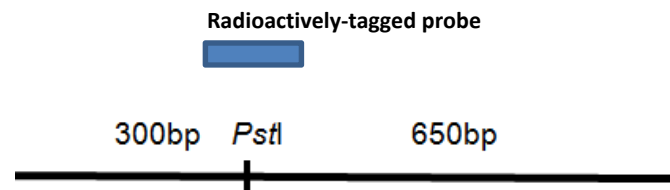
(a) Using the information in Fig. 2.1,

(i) state and explain which restriction enzyme digest should be used to detect the XD15 genetic marker associated with disease X.[3]

- *Pst*I
- Digestion with *Pst*I produced two bands of 300bp and 650bp in affected individuals but unaffected individuals showed either a single band of 950bp or three bands of 950bp, 650bp and 300bp.
- But after digestion with *Eco*RI, all individuals had single 950bp fragments and the affected and unaffected individuals are not differentiated.

- (ii) draw a restriction map of the XD15 genetic marker that is associated with disease X. Indicate on the restriction map the position of the radioactively-tagged probe that would enable visualization of the RFLP fragments.

[2]



- [Size of fragment + labelling of restriction site] -1m
- [Position of probe + label of probe] -1m

- (b) Explain why genetic markers like XD15 can be used to detect the presence of disease-causing alleles.

[2]

- These genetic markers are easily identifiable because they are highly polymorphic
- They are closely/ tightly linked to the disease-causing alleles on the same chromosome, thus unlikely that crossing over will occur /are likely to be inherited together as one unit with the disease-causing allele

The XD15 locus was amplified by polymerase chain reaction prior to gel electrophoresis and southern blot. The DNA sequence of the XD15 locus is shown in Fig. 2.2.

5' - GGATCCATCCCGATCGAAAGCTAGCTAGGATCC - 3'
 3' - CCTAGGTAGGGCTAGCTTTTCGATCGATCCTAGG - 5'

Fig. 2.2

- (c) Design two 7-base long primers for the sequence to be amplified.
 [2]

Primer 1: 5' – GGATCCA – 3'

Primer 2: 3' – TCCTAGG – 5'

- 5' and 3' indicated correctly on both primers- 1m
- Correct sequence of both primers – 1m
- 1m awarded if one of the primer has correct sequence and direction indicated.

- (d) Contrast between the process of PCR and DNA replication that occurs naturally in cells.
 [3]

Basis of comparison)	Differences
1. Nature of primers	DNA replication involves RNA primers while PCR requires DNA primers.
2. Location	DNA replication takes place in the nucleus of the cell while PCR is automated/takes place in a thermocycler.
3. Enzymes involved	DNA replication involves DNA polymerase while PCR involves the enzyme <i>Taq</i> polymerase.
4. Proof-reading	In DNA replication, the daughter strands are proofread by DNA polymerase I but there is no proofreading of daughter strands in PCR by <i>Taq</i> polymerase.
5. Unzipping of template DNA	In PCR, high temperatures are required for denaturing the strands while in DNA replications, the enzyme helicase unwinds the strands.
6. Synthesis of primers	In DNA replication, primase synthesises the RNA primers, but in PCR, the primers are added in/primase is not involved.

[max 3]

- (e) Besides disease detection, RFLP analysis may be used for DNA fingerprinting as well. Explain one difference in the approach employed during RFLP analysis for both processes (disease detection and DNA fingerprinting). [2]

	Disease detection	DNA fingerprinting
Number of DNA probes used	One probe;	Multiple probes;
→ Reason	Only one loci is analysed in disease detection;	multiple loci are analysed in the creation of a DNA fingerprint;
Location where restriction enzymes cut	Restriction enzymes cut within the gene locus	Restriction enzymes cut outside the Variable number of tandem repeat (VNTR) loci
→ Reason	Changes in base sequence changes restriction sites that result in different sized restriction fragments being released when cut with the restriction enzyme.	The DNA polymorphism for this loci arises due to different number of tandem repeats of a particular sequence. Cutting outside the VNTR loci releases fragments of different lengths for each allele. Gel electrophoresis can then separate the different alleles by size.

[max 2: state difference 1m + explanation 1m]

[Total: 14 marks]

Question 3

(a) Plant growth regulators (PGR) such as auxin, cytokinin and ethylene are naturally produced in plants and are important in determination of the developmental pathway of plant cells.

Synthetic analogues that can be mass produced are frequently used in plant tissue culture instead of the natural PGR.

Table 3.1 summarizes the general characteristics and roles of some natural and synthetic PGR.

Type of PGR	Examples	Role
Auxins	IAA(natural) - unstable to heat & light	Promote cell division and cell growth Root initiation (when auxin:cytokinin is high)
	2,4-D(synthetic) – stable to heat & light	
Cytokinins	2iP(natural) – unstable to heat & light	Promotes cell division Shoot formation (when auxin:cytokinin is low)
	Kinetin(synthetic) - stable to heat & light	
Abscisic acid (ABA)	-	Inhibits cell division Maturation of somatic embryos Abscission (i.e. shedding) of plant leaves Seed dormancy Induces stomatal closure to reduce water loss by transpiration
Ethylene	-	Abscission (i.e. shedding) of leaves and flowers Seed and bud dormancy by growth inhibition Fruit ripening

Table 3.1

Prunus lannesiana is an early-flowering cherry (only in spring) in Japan Izu peninsula. It is commonly known as Sakura or Japanese Cherry. Researchers are keen to propagate *P. lannesiana* from sterilized explants by micropropagation due to the advantages that the

technique offers. Hence, a study was made to analyse the concentration of PGR present in the *P. lannesiana* in the four seasons and the results were summarized in Table 3.2.

Conc. of PGR / arbitrary unit \ Season	Spring (Mar-May)	Summer (June – Aug)	Autumn (Sep-Nov)	Winter (Dec – Feb)
IAA (auxin)	15	20	25	30
2iP (cytokinin)	10	15	10	5
ABA	5	5	12	15
Ethylene	12 – 20	18	16	14

Table 3.2

Table 3.3 illustrates the physiological development of *P. lannesiana* in the various seasons.

Season	Spring	Summer	Autumn	Winter
Shoot development	Very Active	Active	Minimal	Nil
Root development	Minimal	Moderate	Active	Very Active
Leaves development	Very Active	Active	Senescence (leaves turning autumn yellow) and abscission	Nil
Flowers development	Short full bloom in early spring, followed by senescence	Nil	Nil	Buds develop but dormant in late winter
Fruit and seed development	Fruit and seed development in late spring	Seed maturation	Seed dormant	Seed dormant

Table 3.3

(i) State one economical limitation of micropropagation. [1]

1. Micropropagation requires sophisticated facilities, sterile laboratory conditions and special nutrient media – high cost to maintain
2. It also requires trained personnel with specialised skills / technical expertise – high cost to train;
3. AVP

(ii) Suggest why synthetic PGR (e.g. 2,4-D and Kinetin) are used instead of natural PGR (e.g. IAA and 2iP) in plant tissue culture. [1]

1. **Lower Cost due to mass production;**
2. **More stable to heat and light;**

(iii) Using the information provided in Table 3.2 and Table 3.3, explain the effect of the change in the auxin:cytokinin ratio from Spring to Winter and vice versa. [4]

1. When season change from Spring to Winter, the **auxin:cytokinin ratio generally increased from 3:2 in spring to 6:1 in winter / by four fold;**
2. which **promotes root growth in winter to improve absorption of nutrients and water** to sustain the survival of plant;
3. When season change from Winter to Spring, the **auxin:cytokinin ratio decreased from 6:1 in winter to 3:2 in spring / by four fold;**
4. which **promotes shoot development in spring, allowing for full bloom of *Prunus lannesiana* in spring for pollination / for photosynthesis for maximal growth / for regeneration of leaves for photosynthesis;**

- (b) The Salmon Genome Project (SGP) is developed to increase knowledge of the biology of Atlantic salmon and aid agricultural breeding of the fish.

- (i) Describe how the SGP serves to increase knowledge of the biology of Atlantic salmon. [2]

1. To **generate the genomic map** of the Atlantic Salmon;
2. To identify, locate and analyse genes and their regulatory sequences;
3. To study patterns of inheritance of these genes;
4. To establish possible evolutionary relationships with other organisms;

In order to produce transgenic salmon expressing salmon growth hormone, sGH, the coding sequence of sGH gene is isolated from a library and cloned, using *Sac* I, into the plasmid expression vector, pBluescript II SK. The plasmid map is shown in **Fig. 3.1**.

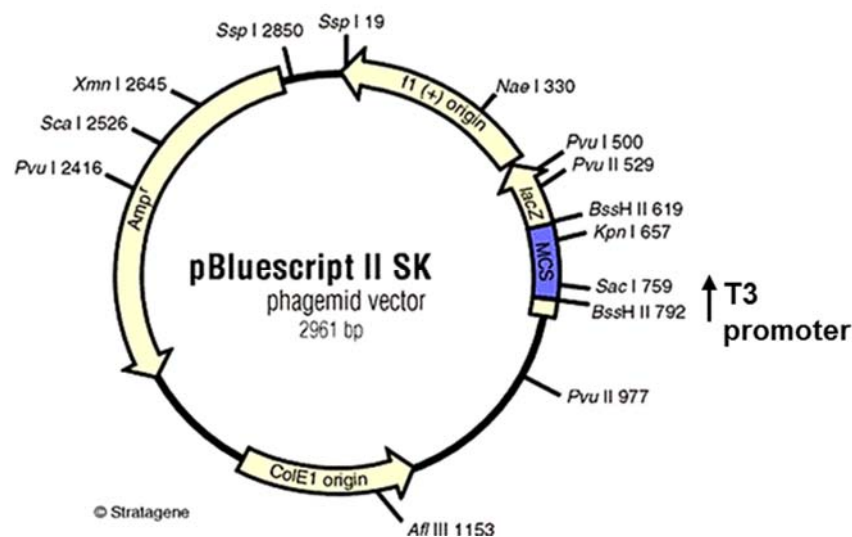


Figure 3.1

- (ii) With reference to figure 3.1, describe the features of the multiple cloning site (MCS). [2]

1. Consists of a **variety of restriction enzyme sites / recognition sites** for **different** restriction enzymes such as *BssH* II, *Sac* I and *Kpn* I;
2. **Found within** the selection marker ***lacZ* gene**;
3. There is only **one / unique recognition site** for **each restriction enzyme** for the insertion of the gene of interest (GOI) into the vector;

Subsequent to insertion of sGH gene into pBlueScript II SK, transformation into *E. coli* cells was carried out and some colonies were obtained. The plasmid DNA was extracted, digested with *Sac* I and the restriction fragments were separated in gel electrophoresis.

- (iii) It was found that the recombinant plasmid with sGH gene inserted yielded no polypeptide. With reference to Fig. 3.1, state and explain a reason for this. [2]

Either:

1. Since ***Sac* I restriction site is within / near T3 promoter**, sGH gene inserted may **disrupt the promoter sequence**;
2. This results in **no expression** / transcription & translation of gene & thus **no polypeptide** is produced;

OR:

3. sGH inserted in the **wrong orientation**, hence **sequence of gene is reversed**;
4. So even though a protein is expressed under the control of T3 promoter, the amino acid sequence and hence the **3D conformation of polypeptide** is **different** from original;

[Total: 12 marks]

Question 4

Cefazolin is an antibiotic that disrupts the synthesis of bacterial peptidoglycan cell by preventing the formation of peptide bonds. It is bactericidal (kills bacteria) and is effective against gram-positive bacteria.

Streptococcus pneumoniae is a bacterium responsible for conditions like pneumonia and bacterial meningitis. It is a gram-positive bacterium which establishes itself as small white colonies. Discs containing cefazolin can be placed on an agar plate containing *Streptococcus pneumoniae*. If the cefazolin has been effective against *Streptococcus pneumoniae*, a clear zone will be seen around the disc as shown in **Figure 4.1**.

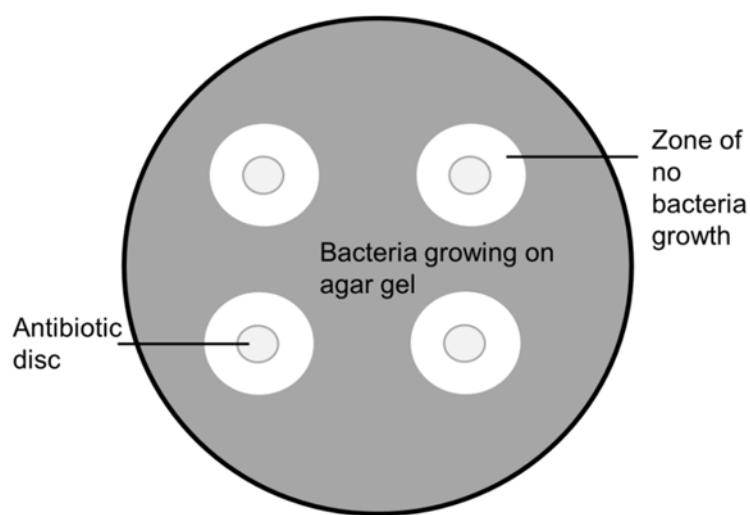


Figure 4.1

You are to plan but not carry out an experiment to investigate the effectiveness of different concentrations of cefazolin on the growth of *Streptococcus pneumoniae*.

Your plan must be based on the assumption that you have been provided with the following equipment and materials:

- Bunsen burner, to enable good aseptic conditions
 - Bacterial culture in nutrient broth
 - Molten nutrient agar
 - Distilled water
 - Sterile 90mm Petri dishes
 - Sterile loops (to plate bacteria onto nutrient agar)
 - 1cm³ pipette
 - Filter paper discs
 - Forceps
 - Vernier calipers
 - 1% cefazolin solution
 - Bactericidal disinfectant for containment of used forceps and pipettes, also to clean work surfaces
-
- Your plan should include:
 - a clear and helpful structure such that the method you use is able to be repeated by anyone reading it
 - an explanation of theory to support your practical procedure
 - an explanation of the dependent and independent variables involved
 - relevant, clearly labeled diagrams, if necessary
 - proposed layout of results tables and graphs with clear headings and labels
 - correct use of scientific and technical terms

- safety measures to minimise any risks associated with the proposed experiment

Total: [12 marks]

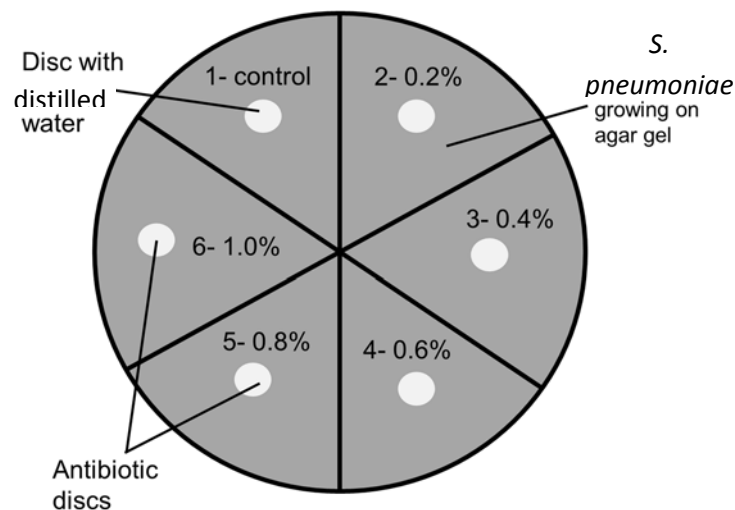
Question 4: Planning Answer

Theory	• Cefazolin is an antibiotic that interferes with the synthesis of peptidoglycan in bacterial cell wall by disrupting peptide bond formation. • This greatly <u>weakens the cell wall</u> and causes the <u>bacterium to lyse or burst open</u>, because of osmotic pressure. As a result, this directly <u>kills the bacteria</u>. • cefazolin is placed in the form of antibiotic discs on the agar gel plated with <i>Streptococcus pneumoniae</i>. The <u>size of the clear zone</u> formed after incubation is a measure of the <u>effectiveness of the cefazolin</u>. Increasing concentration of cefazolin will increase the diameter of the clear zone.	✓ description of scientific reasoning and theory of the method used to measure effectiveness of cefazolin																								
Variables	Independent variables: <u>5 concentrations of cefazolin solution</u> (0.2, 0.4, 0.6, 0.8, 1.0 %) prepared by simple dilution. Dependent variables: <u>Diameter of clear zone / mm</u> Controlled variables (<u>any 2</u>): • Concentration and volume of bacterial culture • Concentration and volume of agar used • Size of filter paper disc • Weight of filter paper	✓ independent , dependent variables and controlled variables																								
Procedure	1. Add 10cm³ of molten nutrient agar to the petri dish using aseptic technique and leave to set. 2. Prepare cefazolin solutions of 5 different concentrations using <u>simple dilution</u> according to the table below. Label the beakers accordingly. <table><tr><th>Concentration of cefazolin solution / %</th><th>Volume of 1% cefazolin stock solution / cm³</th><th>Volume of distilled water / cm³</th><th>Total volume / cm³</th></tr><tr><td>1.0</td><td>10</td><td>0</td><td>10</td></tr><tr><td>0.8</td><td>8.0</td><td>2.0</td><td>10</td></tr><tr><td>0.6</td><td>6.0</td><td>4.0</td><td>10</td></tr><tr><td>0.4</td><td>4.0</td><td>6.0</td><td>10</td></tr><tr><td>0.2</td><td>2.0</td><td>8.0</td><td>10</td></tr></table> 3. Draw lines on the base of the petri dish so that the base is divided into 6 equal parts. Label the sections 1 to 6.	Concentration of cefazolin solution / %	Volume of 1% cefazolin stock solution / cm ³	Volume of distilled water / cm ³	Total volume / cm ³	1.0	10	0	10	0.8	8.0	2.0	10	0.6	6.0	4.0	10	0.4	4.0	6.0	10	0.2	2.0	8.0	10	✓ dilution method
Concentration of cefazolin solution / %	Volume of 1% cefazolin stock solution / cm ³	Volume of distilled water / cm ³	Total volume / cm ³																							
1.0	10	0	10																							
0.8	8.0	2.0	10																							
0.6	6.0	4.0	10																							
0.4	4.0	6.0	10																							
0.2	2.0	8.0	10																							

4. When the gel is cast on the petri dish, use a sterile loop to plate the nutrient broth containing *Streptococcus pneumoniae* onto the agar gel.
5. Prepare a control disc by soaking a sterile disc in distilled water.
6. Using the Bunsen burner, flame and cool forceps to sterilize them.
7. Use sterile forceps to place the control disc in the centre of section 1
8. Prepare the 5 cefazolin discs containing 5 different cefazolin concentrations by soaking each of the discs into the 5 different concentrations of cefazolin solutions prepared from simple dilution.
9. Using flamed and cooled forceps place the disc containing the 5 concentrations into the other 5 labeled sections.
10. The setup of the petri dish in the experiment is as shown below:

- ✓ control

- ✓ method-setup



- ✓ relevant diagram

11. Conduct 3 replicates from steps 1-8 and 2 repeats for all prepared concentration of cefazolin.
12. Seal the petri dish and incubate the petri dish at 30°C for 2 days.
13. Without opening the lid measure the diameter of the clear zone around each disc using vernier calipers.
14. Tabulate the results to show the effect of different cefazolin concentrations on *Streptococcus pneumoniae* growth.

- ✓ 3 replicates, 2 repeats

- ✓ fixed incubation time

Table showing the effect of different concentration of cefazolin/% on the diameter of clear zone /mm

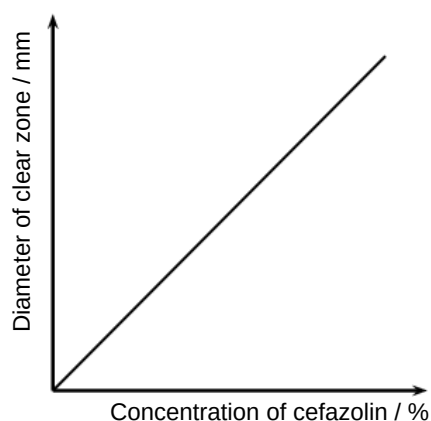
Conc. of cefazolin	Diameter of clear zone /mm			
	Reading 1	Reading 2	Reading 3	Average
1.0				
0.8				
0.6				
0.4				
0.2				
0.0 (Control)				

Average = $\frac{\text{reading 1} + \text{reading 2} + \text{reading 3}}{3}$

3

15. Plot a graph of diameter of clear zone /mm produced against the concentration of cefazolin /%.

Graph of diameter of clear zone /mm produced against the concentration of cefazolin /%.



The increase in the diameter of the clear zones should be directly proportional to the concentration of cefazolin.

✓ method to calculate growth of bacteria

✓ appropriate table and graph

✓ expected relationship between [penicillin] and diameter of clear zone

Risk and precaution	To prevent infection or growth of harmful microorganisms: • Cover all cut or broken skin with a waterproof dressing • Wear tightly fitting disposable gloves and clean laboratory coat • Clean the bench surface with bactericidal disinfectant and use a bunsen burner which creates a sterile environment. Students should work as close as possible to the flame. • Swap any spillages with bactericidal disinfectant. • Proper disposal / treatment of contaminated materials or equipment using sterilizer/autoclave	✓ Safety (Any 1)
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Question 5

Free-response question

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.

Your answers must be in continuous prose where appropriate.

Your answers must be set out in sections (a), (b), etc as indicated in the question.

- (a) Discuss the advantages and evolutionary consequences of using plant tissue culture to propagate transgenic plants. [8]
- (b) Outline the ethical and social implications of genetically modified organisms. [6]
- (c) Explain why gene therapy is opined to have limited success in the treatment of genetic diseases. [6]

Question 5

(a) Discuss the advantages and evolutionary consequences of using plant tissue culture to propagate transgenic plants. [8]

Advantages [MAX = 5]:

1. All transgenic plants contain the desired gene - Using plant tissue culture of a genetically engineered plant cell will give rise to whole transgenic plants that are genetically identical
2. Rapid production - allows large numbers of transgenic plants to be grown from just one or a few stock plants can be achieved; through asexual means;
3. Plant diseases can be avoided - Production of bacteria or virus-free plants; since only meristematic tissue of the transgenic plant is used for propagation **OR** plant cloning method uses dilute sodium hypochlorite for surface sterilization, ensuring bacteria or virus-free plants
4. Transgenic plants can be produced at any time of the year/ not subject to seasonal change; thus plants can be produced out of season and people can buy plants at lower prices than usual;
5. Reliability; and quality control can be achieved - since growing conditions are standardized and optimized
6. Land area needed to grow these transgenic plants through tissue culture is much less; save space;
7. Prevents "loss" of transgene due to sexual reproduction / through gamete formation; maintenance of transgene within entire crop;

Evolutionary Consequences [MAX = 3]:

7. Desirable / beneficial genes are quickly passed on to subsequent generations / ensures integrity and maintenance of genetic composition as the transgenic plants can propagate

rapidly under favourable environment;

8. Lack of genetic variation poses a problem to the survival of a species / unable to respond to unfavourable changes in selection pressure as a result of changing environment;

9. Low evolutionary potential thus no speciation / natural selection cannot occur without variation;

10. Thus there is a high risk of extinction of that species as all individuals are genetically identical, thus equally susceptible to pathogens / succumb to outbreak of epidemics / disease;

11. When transgenic plants are reintroduced into the wild, might cross- pollinate and these might cause production of pesticide/herbicide resistance / generating "superweeds" that will disrupt the ecological balance / ecosystem.

Note: Ans must make reference to transgenic plants, not a generic propagation of plants by plant cloning,

(b) Outline the ethical and social implications of genetically modified organisms. [6]

Definition of GMO – 1m

Organism that has acquired one or more genes by recombinant DNA technology. The genes may or may not be from the same species.

Ethical concerns (Max 3):

1. Exploitation of animals for food (+ any 1 elaborated point)
 - A. Increased use of growth hormone has harmful effects on the health of animals. Eg the use of bovine somatotrophin in dairy cattle increases the risk of mastitis
 - B. Concern whether the animals are biologically capable of withstanding additional stress of increased production of milk, meat and other products
2. Exploitation of animals for medical research (+ elaboration)
 - A. Medical experiments may cause suffering in animals and there are concerns of violations of animals' rights. (eg. Oncomouse)
3. Religious concerns or dietary restrictions (+ any 1 elaborated point)
 - A. Eg. Religious groups are concerned that GM foods might contain genes from animals prohibited by their religion
 - B. Objections to consumption of plants that have been modified to carry animal genes or vice versa by vegetarians.
4. There is concern about the rights of patenting a genetically modified animal or plant. (+ elaboration)
 - A. Companies have sought to patent the transgenic animals or plants that they have developed, however, people argue that patenting animals is unethical as it reduces them to the level of objects.

5. **Labelling of products** on sale to indicate that genetic engineering was involved in their production is **not mandatory** in some countries (+elaboration)
A. this deprives consumers from making an informed choice based on their religious, medical (allergies), personal (vegetarians) backgrounds.

Social Concerns (max 3)

1. Release (accidental or otherwise) of GM animals into the wild may result in GM animals **outcompeting wild types** such that **ecological balance is disrupted**/severe impacts on the food-chain.
2. Introduction of foreign gene(s) may result in production of secondary metabolites that may be **toxic** to animals themselves and/or livestock/humans that consume them.
3. New proteins in GM animals may be potentially **allergenic** to humans that consume them.
4. **Antibiotic resistance genes** may be transferred to bacteria in the gut, increasing the resistance of such bacteria to medicinal antibiotics.
5. E.g., larger transgenic salmon may be preferably selected as mates over smaller wild types, thus **destabilises ecosystem** and hence **threatens biodiversity**;

(c) Explain why gene therapy is opined to have limited success in the treatment of genetic diseases. [6]

1. Short-lived nature
 - Many vectors **do not** allow for the **integration of the normal allele into the target cell genome** / the **rapidly dividing nature of many cells** (i.e. new cells need to be targeted) prevents the effects of gene therapy from being long-lived and stable.
 - As a result, patients have to undergo **multiple rounds of gene therapy**.
2. Immune response
 - The **vector** that is introduced into the tissue might be recognised by the immune system as a **foreign** particle (e.g. due to previous infection) and elicit an **immune response**.
 - As a result, the **effectiveness** of the therapy is **reduced** as the vectors are **destroyed**.
3. Viral vector may regain virulence
 - Even though the in the viral vector is modified to be safe/disease-causing viral gene is removed, the viral vector may **regain or develop virulence**.
 - This causes **harm** in the patient body.
4. Insertional mutagenesis
 - For some viral vectors (e.g. retrovirus), the **integration of normal allele into host genome is random** and may **disrupt host's gene** or regulatory gene sequences/control elements.
 - Disruption of tumour suppressor gene/ conversion of proto-oncogene to oncogene (**one named e.g.**) may cause **cancer**.

5. Multi-gene disorders/disease
 - Many common **diseases** (**one named e.g.** heart disease/high blood pressure/ Alzheimer's disease/diabetes), are caused by the combined effects of multiple genes.
 - Gene therapy is not effective as it is impossible to introduce many normal alleles into target cells at the same time.
6. Large genes
 - There may be a problem finding a suitable vector if the **normal allele** that needs to be delivered into the target cells is **large** (e.g. viral vectors can only accept genetic material of a certain size).
7. Non-dividing target cells
 - Some viral vectors may not be able to infect non-dividing cells (eg. retrovirus). Hence gene therapy may not be suitable for diseases involving some cell types.
OR
If the modified target cell (with the normal functional allele) does not divide and multiply, the introduced gene is lost when the cell dies.
8. Diseases caused by presence of a dominant allele
 - Gene therapy is ineffective when the mutated defective allele is dominant and the normal allele is recessive.
 - The expression of the normal allele will still be **masked** by the dominant defective allele even after being introduced into the target cell genome
9. Problem with controlling the activity of gene expression
 - Modified cells may not synthesize sufficient functional proteins to bring about an improvement.
 - Due to (**any valid reason**): e.g. not all cells taking up the normal allele / cells not expressing the proteins at the right time (eg. in thalassemia disease) / normal allele not expressed because they are integrated into a heterochromatic region of the host cell genome.
10. AVP
[1 mark for each well-elaborated point]

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